SMW – SECURITY SYSTEMS CONSTRUCTION DOCUMENT

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MAIN CONTRACTOR SECURITY EQUIPMENT

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The Following Security Equipment will be supplied and installed by Main Contractor, and should interface with Security Systems that provided by MEP/Security Systems Subcontractor, as below:

- Portable Metal Detector
- Portable X-ray Detector
- Automatic Security Entry Swing Doors with returning alarm back to Access Control System under this specification requirement
- Road HVM Blocker, interface with Access Control System Long Range Card Reader and Access Control Panel provided by MEP/Security Systems Subcontractor for Blocker Opening and Closing, with returning alarm back to Access Control System under this specification requirement
- Security Vehicle Shutter, interface with Access Control System Card Reader and Access Control Panel provided by MEP/Security Systems Subcontractor, for Shutter Opening and Closing, with returning alarm back to Access Control System under this specification requirement
- Fire and Security Exit Door, allow mounting of an magnetic devises sensors provided by MEP/Security Systems Subcontractor mounted on each of the leaf of door.

A. Portable Metal Detector

1. There shall be freestanding walk through metal detecting equipment to scan personnel for metal objects that may indicate restricted items.
2. The General Functionality Requirements
   The Walk Through Metal Detector shall be capable of the following:

   - Detection of magnetic, non-magnetic and mixed alloy metal weapons
   - All functions programmable via built-in keypad and display or RS232 serial connection to remote control device, PC or computer network.
   - Access to calibration and operation features protected by a mechanical lock and two alphanumeric passwords
   - Photocell transit counter and automatic alarm rate calculation
   - Speed of detection up to 45 ft/s (15 m/s)
   - Immunity to electrical and mechanical interference including adjacent high energy equipment such as a dual view x-ray scanner.
   - No initial or periodic calibration.

- External RS232 interface for remote control
- Multi-zone display bar for “height on person” localization
- Signals proportional to the mass of a detected object
- Programmable volume and tone of alarm signal

3. General Installation Requirements
CONTRACTOR shall be responsible for providing complete systems and equipment, to be installed within the Building. The work shall include installation and commissioning of the detector as they relate to security systems, which will be further defined within this document.
4. All software licenses as required for the above in the name of the owner for all systems
5. The construction and installation of the equipment shall provide an environment that is robust, stable, reliable and protected to the highest possible degree.
6. As part of the overall security technical requirements there shall be a quality assurance system to monitor activity, and audit user actions for safety, effectiveness and efficiency.
7. The CONTRACTOR shall be responsible for coordinating installation with the, electrical and all other CONTRACTORs in order to assure proper preparation and installation of the Walk Through Metal Detector.
8. Equipment Specifications
   A. Physical Specifications:
      - Total height: 2255 mm
      - Passage height: 2050 mm
      - Passage width: at least 720 mm
   B. Technical Specifications:
      - Operating temperature: 14°F to 149°F (-10°C to 65°C)
      - Relative humidity: from 0 to 95% (without condensation)
      - System Power: 230 VAC +/- 15% / 50 or 60Hz / .5 Amps maximum
      - Meet security levels specified by NIJ Standard 0601.02
      - Compliant with the Human Exposure Safety Requirements specified by NIJ Standard 0601.02
      - Conforms to the current International Security Standards for Walk-Through Metal Detectors
      - Complies with EC Regulations and International Standards relating to Electrical Safety and Electromagnetic Compatibility (EMC)

   The Walk Through Metal Detector shall meet the following:

- NIJ Standard 0601.02 test kit
- Remote control unit with (20 m) of connection cable
- 5 hour external battery supply and automatic charger

B. Portable X-ray Detector

- Within the Ground Floor there shall be X-ray equipment to determine whether baggage, parcels or personal effects are safe, benign and may be carried within the facility.
  - The X-ray Parcel and Baggage scanner shall be dual view to provide a horizontal sideward and vertical/diagonal downward view of the object under inspection to improve throughput and object discrimination. These two perpendicular views shall provide a complete perspective of complex parcels regardless of their orientation in the X-ray system which eliminates the need for screeners to leave their secured post to reposition and re-scan parcels in high security environments. Remote control unit with (20 m) of connection cable
  - 5 hour external battery supply and automatic charger

- The General Functionality Requirements

  The X-ray parcel/package scanner must have the following features:
  - Remote control workstation.
  - Dual views (horizontal sideward and vertical/diagonal downward)
  - Dual energy levels
  - Provide image display on at least two monitors for each monitor view horizontal sideward and vertical/diagonal downward separately
  - The scanner system shall be provided with emergency stop buttons and safety tunnel covers.
  - The scanner system shall be provided with “System Energized” and “X-Ray on” indicators at both ends of the X-
  - Ray cabinet and on the operator workstation.
  - The system shall include a safety interlock system to prevent X-Ray generation in the event of a critical panel being removed.
  - The scanner shall be capable of functioning as a training terminal.
  - The X-ray parcel/package scanner shall have the following viewing and image analysis features:
    - ➢ Capable of a minimum resolution of 1024 x 768 pixels for each view (Horizontal & Vertical)

- Capable of imaging 38 American Wire Gauge (AWG) un-insulated solid copper wire minimum in free space as measured by the American Society of Testing Materials (ASTM).
- Capable of imaging 27 mm minimum steel penetration as measured by the American Society of testing Materials (ASTM).
- Provided with two X-ray generators operating at anode voltage not less than 160kV.
- Capable of discriminating between Organic and Inorganic materials as measured by the American Society of Testing Materials (ASTM)

  Automatic color coding of materials
  Image Zoom
  Variable Density Zoom
  Capable of viewing previous bag
  Manual / Automatic Image Archiving
  Capable of Multi Energy Imaging (4 color)
  High Penetration X-ray Generator
  Enhanced performance X-ray
  Density Threat Alert
  Variable Edge Enhancement
  High/Low Penetration
  Capable of variable Gamma
  Capable of inverse Video
  Pseudo Color display
  Organic/Inorganic Stripping
  Capable of Black and White Viewing
  Variable Color Stripping

- Operating system

Operating System Software for the X-Ray scanning shall have the following features:

- The control and operating system software for the X-Ray scanning equipment shall be an integrated system designed and provided by the X-Ray scanning equipment manufacturer.
- The control and operating system software shall provide a simplified set of controls intended for use by security personnel consisting of only features determined necessary for proper scanning operation.

- The control and operating system software shall provide controlled access (password protected access) to the operation controls of the X-Ray scanning equipment.
- The control and operating system software shall provide multi-level controlled access (password protected access) to all calibration, maintenance and logging functions by authorized technical personnel.

- General Installation Requirements
  - Contractor shall be responsible for providing complete systems and equipment, to be installed within the Scanning Building. The work shall include installation and commissioning of the X-ray parcel/baggage scanning system as they relate to security systems, which will be further defined within this document.

  - All software licenses as required for the above in the name of the owner for all systems
  - The construction and installation of the equipment shall provide an environment that is robust, stable, reliable and protected to the highest possible degree.
  - As part of the overall security technical requirements there shall be a quality assurance system to monitor activity, and audit user actions for safety, effectiveness and efficiency.
  - The Contractor shall be responsible for coordinating installation with the, electrical and all other contractors in order to assure proper preparation and installation of the X-ray parcel/package scanner.

- Equipment Specifications

A. The X-ray parcel/package scanner must meet the following physical specifications:
  - The scanner shall not exceed: Length: 2880 mm, Height: 2000 mm, Width: 1430 mm
  - Tunnel size shall be 640 mm (W) x 430 mm (H)
  - The scanner shall be mounted on heavy-duty castors for ease of movement.
  - It shall be fixed in place by means of adjustable jacking feet (Nominal height +25.4 mm adjustments)

B. The X-ray parcel/package scanner must meet the following technical specifications:

- System Power: 230 VAC +/- 10% / 50 or 60Hz / 7 Amps maximum
- Conveyor speed shall be 0.2 m/sec minimum
- Conveyor load shall be165 Kg minimum evenly distributed
- Operating Temperature range: 0°C to 40°C
- Operating Relative Humidity range: 5 to 95% non-condensing
- Comply with all applicable international health and safety regulations including USA FDA for cabinet X-ray systems (Federal Standard 2.1-CFR 1020.40), the Health and Safety at Work Act 1974-Section 6, amended by the Consumer Protection Act 1987 and Guaranteed even for high speed films up to ISO 1600 (33 DIN)
- The unit must comply with all ruling international safety regulations such as the CE, German TUV, and Swiss SEV.
- The Xray can exhibit images in the Classic 4-color and the new proprietary Spectrum 4-color (SP4) option providing superior image, allowing improved security by quick and accurate identification of threats and increase in throughput. The display unit should be at least a 17inch LCD Monitor

C. Non Powered Roller Conveyor

Two Non Powered Roller Conveyors mainly transport cartons or packaging products are provided in either side of the X-ray detector, Should be provided by the Main contractor.
The size of the conveyor should be maximum of 800mm width, 830mm height and 1000mm long. The height should be merged with the X-ray detector's conveyor height.

It should be easy to operate, easy to use, can be moved and handled by two persons. Roller conveyor per square meter can bear at least 100KG weight loading, the diameter of the roller is about 38mm.

It is made of high quality stainless steel and is very wear-resistant. Also can be customized according to the need of the corresponding conveyor.

Non Powered Roller Conveyor Products features :

1) It has the characteristic of long-distance transport, large volume and continuous transport.

2) It has low possibility for abrasion and brush, the production efficiency is high.

3) Simple structure, easy maintenance

A Sample Non Powered Roller Conveyors

C. Automatic Security Entry Swing Doors

C.1 GENERAL REQUIREMENT

1. The Automatic Security Swing glass door located at the Entrance Lobby of the Data Centre Main Lobby should be burglar-resistant.

2. The automatic Swing door with RC3 Security standard must be TÜV-tested for conformity with EN 16005 and DIN 18650. The Swing door drive unit in combination with RC burglar-resistant profile system must be fulfils European break-in prevention standards EN 1627 - EN 1630. The system as a whole must be tested for conformity with EN 16361, corresponding to SIA 343.

3. The Swing door shall be elegant integration, into the Swing door drive unit, of all actuating components, sensor elements and leaf profiles. It should contain Digital control unit with touch-screen display. An electronic control unit with an automatic reversing and stopping mechanism provides gap-hazard detection. The door is suitable for connection via a BUS configuration or conventional wiring system. Noise-insulated/vibration-damped drive system. Low-noise

toothed belt. Running carriage with counter-pressure rollers, height/depth-adjustable. A series of metal plates and bolted-in elements in the drive case is designed to prevent attempts to break in by levering the door open. Drive case height 140 mm.

C.2 DRIVE CASE COVERING

Standard: Hinge-type aluminium drive unit covering designed for easy service and maintenance access, with mechanical hold-open system and drop prevention.

C.3 RC LEAF PROFILE SYSTEM

All-round frame made of reinforced aluminium profiles. The upper leaf profiles are integrated in the drive unit. Main closing edge uses “groove-and-slot” principle. The robust, on-board locking latch with slide element engages with the flush-fitting bottom guide rail. The bottom locking latch also reinforces the corner of the door leaf frame. Vertical interlocking between the Swing door leaves and side panels increases safety in the closed position, ensures stability and provides protection from draughts. Wet glazing with state-of-the-art adhesive technology is required. IV-VSG insulating laminated safety glass, 24 mm thick. An IV-VSG P5A double insulating glass, 24 mm thick shall be used.

C.5 SURFACE TREATMENT

The system shall be with surface treatment of powder coated in RAL colours or aluminium in natural anodised finish E6/EV1.

C.6 BOTTOM GUIDE RAIL RC

With full-length rail embedded in floor, for reliable operation and increased resistance to wind pressure and forced opening. The door leaves are guided over the entire width and the guide rail shall be replaceable.

C.7 EMERGENCY OPERATING MODE

If the power supply fails, batteries ensure operation for a further 60 minutes. The system continuously monitors the battery's charge level to ensure that it will work in an emergency, and duplicate all these alarm to Security Access Control and Alarm System through normal close contact. The final door movement (opening or closing) in the event of low charge state can be programmed.

C.8 RC locking mechanism

Multi-point locking mechanism installed inside the door leaf. Four robust, securely encased locking bolts made of steel offer high protection against break-ins. The locking mechanism shall be automatic and able to be manual unlocking mechanism using cylinder

The system shall be able to monitoring of the door leaf position and lock status by TUWE (VdS), for connection to an alarm system.

C.9 SENSORY SYSTEM AND USER PROTECTION

The system shall be included integrated opening/safety sensor fitted externally in the cover or carrier profile. It should include Motion detector and scanning safety barrier for the reliable monitoring of opening and closing movements.

The system shall be included integrated safety sensor fitted inside the drive case. It should include Scanning safety barrier for reliable monitoring of the secondary closing edges during opening.

C.10 CONTROL ELEMENTS

The system must include a digital control panel with illuminated touch-screen display, for selecting the door functions, adjusting all door parameters and displaying the operating data. The illuminated touch-screen display will be provided in G/F Security Guard Room.

The Doors should provide with an Normal close contacts status signal should also be provided by this contractor to indicate the permanent Closing status of the doors to the Access Control System provided by MEP/Security Systems Subcontractor. Detail logic should be coordinated between MEP/Security Systems Subcontractor and this Main Contractor. If the shutter is malfunction due to power overload and/or Power supply failure, an normal Closed contact alarm should return back to Access Control System under this specification requirement.

All the above interfacing should be integrated in an interface marshalling box provided by this contractor in G/F Security Guard Room.

C.11 TESTING

The Door design must fully comply with CE regulations and an actual full scale crash test must have been carried out by a qualified independent testing agency with the shutter affording protection after the impact.

The impact testing must have been carried out in accordance with RC3 Security standard and must be TÜV-tested for conformity with EN 16005 and DIN 18650.

The main contractor should also test the interface as above specification requirement with the MEP/Security Systems Subcontractor, to satisfy the indication requirement in MEP/Security Systems Specification.

D. Road HVM Blocker

a. Road blocker

The Scope shall include the Supply, Install and Co-ordination of the 2 Road blockers for Data Centre. The Tenderer shall be Responsible for Road Blocker Installation, Co-ordination and care until project Completion and Handover.

b. Blocking Segment

b1. Segment Construction

The blocking segment shall be constructed of heavy steel section with mild steel skirts and covers. The road plate should be an anti-slip plate to withstand a calculated axle load of 30 tones. The blocking segment shall be secured into a sub-surface mounting frame and will be rotated through 43 degrees to secure the roadway with the rotation shaft secured through Nylocross self lubricating bearings.

On impact, the forces exerted on the blocking segment should be partially absorbed by the structure with the balance being transmitted through the substantial frame and into the re-enforced foundations. The segment should be designed to absorb / withstand impacts from both US and European manufactured vehicles taking into account the varying weight distribution of both styles.

Folding skirt must be fitted to the blocking segment to provide protection against the ingress of debris into the Road blocker as well as acting as an anti-crush safety barrier.

The mounting frame should be fitted into a fully reinforced foundation which will have a maximum overall depth of 300mm with drainage points to connect to suitable soakaway or storm drainage. If required, a sump pump may be provided. Specialized shallow mounted roadblocking system for use in circumstances where deeper foundations are not feasible.

b.2 Segment Height

The height of the segment when in the closed (raised) position, as measured from the top of the mounting frame, will be a minimum of 1000mm to minimize the possibility of site penetration and to ensure that higher chassis vehicles are restrained.

High Tensile Steel chains are to be incorporated to stop the blocking segment from over-rotating.

b.3 Segment Width

The width of the blocking segment will be 2,000mm.

b.4 Finish

The segment and mounting frame are to be finished with an anti-corrosion paint in black with yellow diagonal stripes over-painted on the segments front face and road plate.

c. Hydraulic System (HPU)

c.1 Operation

The HPU will consist of a heavy duty motor driving a hydraulic pump which will actuate, via a manifold and set of electrically operated valves, a pair of hydraulic rams. The blocking segment will be driven both up and down by the double acting hydraulic rams to ensure positive action at all times without reliance on gravity for operation.

The HPU will have the facility for a hydraulic accumulator to be added to the system to allow a significantly increased raise speed in case of emergency.

c.2 Hydraulic rams

The double acting hydraulic rams fitted to the blocking segment can be fitted with anti-burst valves as to maintain security if damage is sustained to the hydraulic hoses.

For safe maintenance, the hydraulic rams must be accessible from the road surface when the Road blocker is in the fully lowered position.

c.3 Limit Switches

Proximity limit switches will be fitted to provide raised and lowered signalling to the control system. The switches will be M18 sized, inductive type with no moving parts and have a minimum IP rating of IP68.

c.4 Motor

The heavy duty motor used in the HPU will be a 3ph, 380-415v unit with a power rating sufficiently sized to allow for continuous operation (100% duty cycling). The motor should be protected by a thermal / magnetic trip device mounted in the electrical control panel.

c.5 Hydraulic Reservoir

The hydraulic oil will be contained in a steel reservoir which is to be sized to allow sufficient oil cooling necessary for 100% duty cycling of the blocker.

If the system is to be used in extremely low temperatures then an immersion tank heater can be added to maintain the minimum oil temperature for operation.

c.6 Power fail conditions

A hand pump / release will be provided to enable the manual raising and lowering of the blocking segment in the event of electrical power failure.

c.7 Casing

The HPU is to be fitted into a steel cradle which will have lifting and bolt down points, the cradle will come complete with an outer cabinet to give protection against the elements.

The cabinet will have fully lockable and removable full length doors to both the front and rear of the cabinet for ease of access. Vents will be fitted into the cabinet to allow good air circulation maintaining the ambient temperature.

d. Control System

d.1 Main Processor

The HPU will be controlled by a central programmable logic controller (PLC) which will accept inputs from the access control system, blocker monitoring equipment and hydraulic pack and output signals to the HPU control valves, back indication system and external signalling. The PLC controller shall be sized to suit site requirements but should have 16 inputs and 14 outputs as a minimum. The initial PLC programming shall be to suit specific client and site requirements however reprogramming of the system must be easily undertaken and password protected access to the program must be provided to the client with the relevant program ladder diagram.

All relays will be properly mounted and all interconnecting cabling must be in suitable containment running to terminal strips.

d.2 Voltage

The main system input voltage is to be 380v-415v 3phase 50-60Hz supply with the control system operating at 24V SELV provided from an internally mounted power supply.

d.3 Casing

The control system will be housed in a general purpose IP65 rated housing with a power isolation switch mounted externally for safety. The housing will be located inside the main HPU cabinet and should give easy access to all electrical components for connection, maintenance and programming.

d.4 Traffic Lighting Control System

The Traffic Control Systems shall be used along with thefollowing:

Electrolier: The complete assembly of pole, mast arm, luminaire, ballast, and lamp.

Luminaire: The assembly which houses the light source and controls the light emitted from the light source. Luminaires consist of hood (including socket), reflector, and glass globe or refractor.

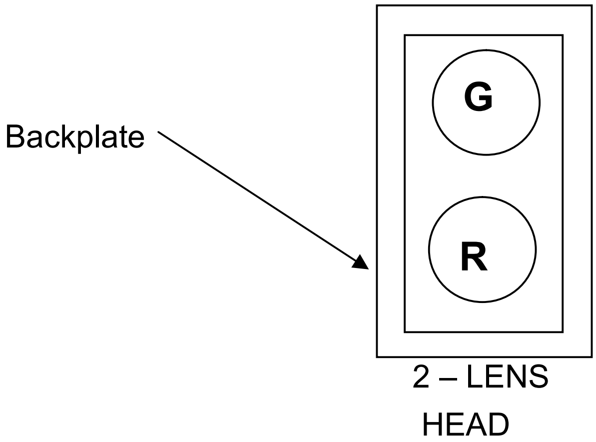
Lighting Standard: The pole and mast arm which must support the luminaire.

Controller Unit: The solid-state device conform to standards for microprocessor-based, solid state, digital timed, signal controller providing specific phases of signal control, internal preemption, time base coordination, internal time-of-day programming, and data base management by a computer.

Controller Cabinet: A cabinet constructed, wired and well equipped.

Controller Assembly: The controller cabinet, controller unit and the equipment described above. The controller assembly shall operate various traffic signal devices to provide right-of-way, clearance and other indications with duration and sequence as determined by preset programming.

Anchor bolts apply to Luminaire poles and anchor rods apply to Signal poles.



e. Access Control

e.1 Remote Control Panel

Each Road blocker will come with its own remote control panel which will be comprised of push buttons to raise, lower and emergency stop the equipment All Push button should be provided with a protection cover to avoid trigger the shutter accidentally. The blocking segment must be able to instantly reverse on command. Two Remote control panels should provide with located at Security Portable Guard House and Security Guard Room at Ground Floor at Data Centre respectively..

e.2 Emergency

Fast raise If the system is fitted with an emergency fast raise circuit, then the control panel will include a panic button which will be larger then the normal controls and will self lock on activation (key reset required), the circuit will by-pass all safety systems and raise the blocking segment within 1 second. An normal Closed contact alarm should also return back to Access Control System provided by MEP/Security Systems Subcontractor.

e.3 System Interfacing with Access Control system

The control system will be capable of accepting inputs from every major type of access control including but not limited to – Swipe card readers, proximity card readers, inductive loop systems, RF transmitter equipment and Long Range card readers provided by MEP/Security Systems Subcontractor. The blockers should interface with Access Control System Long Range Card Reader and Access Control Panel provided by MEP/Security Systems Subcontractor, for Access Control to control the each Blocker Opening and Closing , Normal close contacts status signal should also be provided by this contractor to indicate the Opening and Closing status of the blockers to the Access Control System provided by MEP/Security Systems Subcontractor. Detail control logic should be coordinated between MEP/Security Systems Subcontractor and this Main Contractor. If the shutter is malfunction due to power overload and/or Power supply failure, an normal Closed contact alarm should return back to Access Control System under this specification requirement.

All the above interfacing should be integrated in an interface marshalling box provided by this contractor in G/F Security Guard Room.

The system must be able to interface with other equipment (by other manufacturers) to create an interlock.

f. Performance

f.1 Manufacturers Experience

The manufacturer of the Road blocker will have a minimum of 15 years experience in the manufacture of this type of equipment and must be a member of a recognized Professional Trade Association.

f.2 Testing

The Road blocker design must fully comply with CE regulations and an actual full scale crash test must have been carried out by a qualified independent testing agency with the Road blocker affording protection after the impact. The test will have consisted of the impact of a roadworthy and fully laden vehicle weighing 7,500Kg (16,500 pounds) travelling at 80 Kph (50 mph) into the security face of the Road blocker.

The impact testing must have been carried out in accordance with PAS 68 and exceed the DoS test standard SD-STD-2.01 Rev A class K12/L3.

In addition, the Road blocker will be able to cause sufficient damage to a 7500 kg vehicle travelling at both 50 kph and 80kph so as to destroy the front suspension and main drive train of the vehicle rendering it inoperative. The Road blocker will be able to fully working after the impact test, which allows protection from second wave attack and ability to open site, to allow access to emergency vehicles

The main contractor should also test the interface as above specification requirement with the MEP/Security Systems Subcontractor, to satisfy the indication requirement in MEP/Security Systems Specification.

f.3 Speed of operation

Standard operation speed will be between 4 and 8 seconds for either raising or lowering, with the raise time being reduced to less than 1 second when the emergency fast raise system is activated.

In normal operation the Road blocker shall be capable of operating at 225 cycles per hour (100% duty cycling at 8 second settings) and must have been satisfactorily factory tested in a continuous run of 1,800 cycles.

g. Quality Assurance

g.1 Equipment Testing

The manufacturer will have fully tested the Road blocker, HPU, Control System and Access Control equipment prior to despatch. These tests will be fully traceable to each unit despatched and must be transparent. The QA testing will include dimensional checks, workmanship quality and finish as well as full operational testing. Once fully tested, the Road blocker will be fitted with a nameplate bearing the manufacturers details, serial number and test date.

The manufacturers quality system must be certified to ISO 9001.

E. Security Vehicle Shutter

1 General Requirement

1. The Security Vehicle Shutter located at the Car Park Entrance of the Data Centre should be burglar-resistant.

2. The Security Vehicle Shutter with LPS1175 SR3 Security standard must be TÜV-tested.
3. The shutter shall be elegant integration, into the shutter drive unit, of all actuating components, sensor elements and leaf profiles. It should contain Wind – lock guides SR3 strengthened guides with twin brush that can tolerate Wind Class 5.

2 Drive Case Covering

Standard: Hood and Motor cover galvanised designed for easy service and maintenance access, with mechanical hold-open system and drop prevention for future access under CLP (China Lighting) Requirement. Motor should be 415V direct drive or chain drive with safety brake.

3 Speed

STANDARD APPROX. 100MM PER SECOND

4 Surface Treatment

The system shall be with surface treatment of Galvanised or powder coated in RAL colours.

5 Usage Frequency

Support maximum 12 operations per hour.

6 Safety

The Shutter should support hold up to run or Safety Edge and Safety Photocell or Light Curtain, duplicate all these alarm to Security Access Control and Alarm System..

7 Curtain Specification

The Shutter should provide with Security proof Curtain which at least SR3 Twin 18g lath in galvanized steel.

8 Operation mode and Interfacing with Access Control system

The Shutter should provide with Push button on Control panel located at Security Portable Guard House and Security Guard Room at Ground Floor at Data Centre. All Push button should be provided with a protection cover to avoid trigger the shutter accidentally. The Shutter should interface with Access Control System Card Reader and Access Control Panel provided by MEP/Security Systems Subcontractor, for Access Control to control the Shutter Opening and Closing , Normal close contacts status signal should also be provided by this contractor to indicate the Opening and Closing status of the shutter to the Access Control System provided by MEP/Security Systems Subcontractor. Detail control logic should be coordinated between MEP/Security Systems Subcontractor and this Main Contractor. If the shutter is malfunction due to power overload and/or Power supply failure, an normal Closed contact alarm should return back to Access Control System under this specification requirement. All the above interfacing should be integrated in an interface marshalling box provided by this contractor in G/F Security Guard Room.

9. Testing

The Shutter design must fully comply with CE regulations and an actual full scale crash test must have been carried out by a qualified independent testing agency with the shutter affording protection after the impact.

The impact testing must have been carried out in accordance with LPS1175 SR3 Security standard

The main contractor should also test the interface as above specification requirement with the MEP/Security Systems Subcontractor, to satisfy the indication requirement in MEP/Security Systems Specification.

F. Fire and Security Exit Door

1.General

The Staircase Fire and Security Exit Doors located at the Ground and First Floor of the Data Centre Main Lobby should be highest security against forced entry, ballistic and blast attacks, fire and smoke protection, durable and suitable for economic continuous operation. As layout plan indicated, the doors involved include both single or double door

The Door should allow mounting of an Electronic monitoring components magnetic devises sensors and a push bar supply and installed by MEP/Security Systems Subcontractor. It should be mounted on each of the leaf of door. Coordination on mounting method should be merging on site between main contractor and MEP/Security Systems Subcontractor

2.Tested and certified on the Fire and Security door

- Forced entry resistance According to DIN EN 1627-1630 - single door / double door up to up to RC4.
- Fire resistant Classified according DIN EN 13501-2, single and double glazed door up to EI90 (T90), tested on both door faces. Tested according British Standard BS 476 Part 22 with local fire regulation requirement
- Smoke resistant Classified according DIN EN 13501-2: C5 S200, C5 Sa (RS)
- Impact resistance According to DIN EN 13049 Depending on the type of door up to level 5,
- Life cycle test Suitable for frequently used entrances and exits. According to EN 1191 200.000 cycles

3. Performance at a glance :

- Tested as a complete element including glazing, hardware and wall connections in various configurations.
- Door leaf construction: depending on the security level, the door leaf thickness is 72 mm. The door leaf consists of an internal extremely torsion-resistant and patented construction of thick-walled profiles, which allow the integration of all security reinforcements and inserts and stiffen the door leaf under extreme stress. All security components inside the door leaf are firmly connected with the whole construction. In the lock area, additional reinforcement plates are welded on to stabilize the lock.
- Door rebate: standard version 3-sided circumferential, alternative version 4-sided circumferential. The security requirement the door rebate thickness is at least 30 mm. This thickness prevents break in attempts with typical burglary tools as well as attacking the locking points with large, electrically operated tools.
- Floor seal: retractable floor seal, individually designed thresholds.
- Door frames standard version: corner frame. The door frame consists of roll-formed galvanized steel. In the frame profile is a groove for the circumferential sealing profile and another slot for the fixing of flush-fitted strike plates and, if necessary, for the electrical fitting components. All security-relevant machined openings, for the lock, magnetic or bolt

contacts, security bolts, electric strike, in the frame are correspondingly reinforced and on the wall side covered by steel boxes.

- Alternative door frames: block frame and various edge and counter frames.
- Door hinges consist of roll-formed 6 mm thick, 3-part steel components, connected by double ball bearing and hardened hinge bolts. The hinge pin is screwed in a concealed position to prevent removal. The hinges are designed for high swing door weights of up to 1,000 kg, for sliding doors the permissible weight is up to 5,000 kg. In the area of the hinges the frame is reinforced with thick-walled angle steel profiles to transfer the weight of the leaf via the frame into the wall construction.
- Depending on the security level, additional security bolts are installed in the rebate of the hinge side to prevent lever out attempts. They serve also as counter-pressure bearing to prevent lateral displacement of the door leaf.
- Tested with a wide range of locking systems: mechanical, electromechanical or motorized locks, single or multiple locking, self-locking systems and high security robust multipoint vault locking system.
- Steel doors tested with emergency exit locks according to EN 179 and panic exit locks according to EN 1125 up to forced entry resistant level RC52.
- Tested as inward or outward opening door versions, opening angle up to 180°.
- Electronic monitoring components magnetic devises sensors and a Push bar provided by MEP/Security Systems Subcontractor should ensure be integrated to the door by main contractor. The main contractor should also test the interface as above specification requirement with the MEP/Security Systems Subcontractor, to satisfy the indication requirement in MEP/Security Systems Specification.

G. Routine Operational Service Maintenance

G.1. General

a) Tender documents are to include a 12-month comprehensive service maintenance contract that incorporates all parts, labor and 24-hour emergency calls services, with a maximum response on site time of four hours.

b) Proposal for future maintenance for nominal <u>five-year Maintenance and Service</u> shall be sought in the **Schedule of Pricing (SOP)**. The scope of work during DLP period and each five-year Maintenance and Service are listed as enclosed.

c) It is a requirement that the installation shall be visited a minimum of three times a year (at four-monthly intervals) for the purpose of routine preventative maintenance and servicing.

G.2. Scope of Work

a. The contractor shall provide the preventive and corrective maintenance services for Security systems
b. The contractor should hold a valid security company license under the Security and Guarding Services Ordinance (Cap. 460)
c. The contractor should provide all necessary material, tools, equipment and qualified labours to carry out the maintenance for the entire Security System throughout the DLP/ Maintenance and Service period.
d. The contractor is responsible for the Preventive Maintenance Services of the entirely Security system throughout the contract period and submitting a detailed report to the Employer for each preventive maintenance services. The Preventive Maintenance work shall be carried out at normal office hour: from 9:00am to 6:00pm on Monday to Friday & 9:00am to 1:00pm on Saturday, except Public Holiday.
e. The contractor should submit planned maintenance program, maintenance log sheet, system checklists/method statement to show the competence of the maintenance when submitting the tender.
f. The contractor should submit monthly report which includes all incident or request information and summary of parts used each month
g. The contractor should have a monthly meeting with employer to review outstanding incidents and discuss the planned works.
h. The contractor should manage the faulty parts such that he/she can locate the parts if requested by employer.
i. The contractor should keep all the faulty parts unless he/she get approval from employer
j. The contractor should assist employer to maintain inventory and asset
l. The contractor should verify the backup data. And make sure the system can restore to latest status by using the backup data.
m. The contractor should submit detail reports including log sheets, checklists, defectives findings, rectification, method statements,

recommendation, etc. within 14 days after each maintenance work completion and repairing services on all equipments according to the equipment list and maintenance schedule, otherwise payment will not be made.

n. All spare parts or replacement works should comply with the safety requirement of relevant Code of Practice or statutory requirements.

o. During the contract period, the contractor must provide the competent staff to check and repair the systems and equipments as stated at the equipment list in accordance with the following terms and conditions:

a) Attend the emergency call outs which are available twenty-four (24) hours a day, seven (7) days a week. The response time (i.e. On-site) must be within two (2) hours upon any major system breakdown.

b) All rectification works shall be completed within the time period according to the severity of each job. Incident Report with detail explanation of the root cause of the rectification shall be submitted by the contractor within (24) hours upon any breakdown of system/ equipment.

p. If there is any work which may cause disturbance to the occupants of the Data Centre, the contractor has to carry out the work outside normal office hours or as instructed by the Facility Management. The incurred cost for that event is included in this tender.

q. The Tenderer is advised to visit the site, understand the equipment of Security systems in the Data Centre as well as any restriction.

G.3. General Specification of Maintenance and Service

Proposal for future maintenance for nominal five-year Maintenance and Service shall be sought in the **Schedule of Pricing (SOP)**..

a. During the contract period, the Employer reserves the right to change, repair, rewire or re-install the equipment or system which is carried out by third parties, the Contractor shall be responsible for maintaining the Entire System to normal operation after these works at no extra cost.

b. The Contractor shall report to the Employer for any equipment or parts in which become not available or obsolete. The Contractor shall propose the breakdown costs and specifications of the similar substitutes or system upgrade works to the Employer within a reasonable time.

c. Provide backup technical support including workshop repair for defective component, system reliability test, failure analysis & review, alternative spare sourcing & delivery, and refreshment training

d. Every 3-month interval preventive maintenance service for the following systems shall be carried out and completed with testing reports:

- Security Vehicle Shutter
- Road HVM Blocker

The contractor has to well plan this every 3-month interval maintenance work for each system in building over the whole contract period of time. (i.e. four times of this maintenance work for each system in contract period.) And submit report with detail information such as model number, serial number, location etc. Any software license and firmware repair, upgrade should be recorded in the report

d. Every 6-month interval preventive maintenance service for the following equipment in the Electronic Access Control and Alarm system shall be carried out and completed with testing reports:

- Portable X-ray Detector in the Electronic Access Control and Alarm system
- Portable Metal Detector in the Electronic Access Control and Alarm system
- Automatic Security Entry Swing Doors in the Electronic Access Control and Alarm system

The contractor has to well plan this every 6-month interval maintenance work for each system in building over the whole contract period of time. (i.e. two times of this maintenance work for each system in contract period.) And submit report with detail information such as model number, serial number, location etc. Any software license and firmware repair, upgrade should be recorded in the report

e. Every Monthly preventive maintenance service for the security systems shall be carried out and completed with Performance/functional checking of the security systems, ensure all workstations can connect to server, receive alarms, video images and check event log. Submit report by the end of each month with detail information and maintenance summary for various components & equipment. Any software license and firmware repair, upgrade should be recorded in the report.

f. Every Comprehensive maintenance with corrective services for the security systems shall be carried out and completed with System and software maintenance, retrofit, debug, bug analysis, system configuration, also necessary system restore, system recovery & upgrade including operating system. Troubleshooting for the software application and hardware connectivity in the Security Systems.

Submit report within 3 days for each corrective service. Detail information including root cause, parts replacement, log summary, preventive measure etc. shall be recorded. Also, fault rate record shall be reported in monthly report. Any software license and firmware repair, upgrade should be recorded in the report.

G.4. Network Security and Patch Management

a. Contractor shall stringently control and set password to protect the system. Contractor shall also configure the firewall, if available, to allow only legitimate network traffic from external users to the protected networks.
b. For those operation system or software vulnerabilities which are announced by the security advisories of vendors or official website, the contractor shall perform the patching process in a systematic and controlled way.

G.5. Service Level of Rectification Work

The contractor shall follow the Service Level for Rectification Work to ensure all problems are rectified within the specified timeframes when a problem occurs. The course of action to be followed shall be determined by the severity Level of the problem. All problems reported shall be classified and resolve in accordance with the severity level codes. The following shows a definition of each of the four problem severity levels.

Severity Level	Response to on site time	Resolution Time	Remark
1	1 hour (Office Hour); 2 hours(non office hours, public holidays)	8 hours	(90% of Severity ONE problems/incidents shall meet this target)
2	4 hours (Officer Hour)	2 days	(90% of Severity TWO problems/ incidents shall meet this target)

Severity Level	**Description**
1	Major impact on Security services; the problem is of major impact and highly affects the Security Operations of the Data Centre. Full system and/or critical/core services are down impacting the Data Centre ability to provide its Security services in the development
2	Failure or problem that only affects individual areas. The system and/or critical/core services with limitation that are not crucial to the overall operation

G.6 SOFTWARE LICENSE

a) Contractor shall be responsible for obtaining all software user licenses for development and runtime systems as necessary.
b) Application software written specifically for The Employer under this Contract shall remain the property of The Employer and shall not be disclosed, copied or distributed at any time to a third party without The Employer’s explicit written consent.

G.7 TRAINING

The Security Systems Specialist installer shall impart training, to **Qualified** representative’s operating and maintenance personnel, for operating, monitoring and maintenance of the Complete Security Management System installed, utilizing the purpose made “Operation and Maintenance Manuals” and “As-Built” drawings of the project.
While terrorist threats and natural disasters are deemed newsworthy, the activities that protect facilities are often considered mundane. Nevertheless, all the technological devices and written procedures in the world will be ineffective if the devices are not used properly and the

procedures are not executed properly. Through well-conceived, well-executed security training programs, personnel can be better prepared to prevent incidents from happening, respond properly to incidents that do arise, and contribute to recovery efforts more effectively. Without appropriate training, personnel are more likely to contribute to security risks accidentally.

a) Training manuals and training aids should be provided for each trainee, and 4 copies provided for archiving at the company site, Security Designer, Landlord and Consultant.

b) The training manuals should include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. The contractor should furnish all audio/visual and other training materials and supplies.

c) When the contractor presents portions of the course by audiovisual material, copies of the audiovisual material may be delivered to the customer in the same media used during the training sessions.

d) The contractor should also recommend the number of days of training and the number of hours for each day. Approval of the planned training content and schedule should be obtained from the company at least 30 days before the training.

e) All personnel giving instruction should be certified by the equipment manufacturer for the applicable hardware and software being installed.

f) The trainers should have experience in conducting the training at other installations should be approved by the company.

G.8 System Administration Training

a) This training focuses on determining and implementing system operational parameters and making any necessary operational adjustments.

b) The first training class should be scheculed so that it is completed about 30 days before factory acceptance testing (if conducted) or site acceptance testing.

c) A second training class should be conducted one week before the start of acceptance testing, and the system administrators should participate in the acceptance tests and reliability testing.

G.9 System Troubleshooting and Maintenance Training

a) This training focuses on the internal workings of the Security system so that the user can troubleshoot and repair most problems. Topics in this class include system networking communications and diagnostic; device configurations and programming; controller setup, wiring and diagnostics; software troubleshooting; and device programming.

b) The system maintenance training should be taught at the project site about two weeks prior to reliability testing, and these students should participate in the reliability test. The training should include but not limited to the following:

- Security layout of the field equipment
- Security system schematic of each sub system
- Troubleshooting and diagnostic procedures
- Repair instructions
- Preventive maintenance procedure and schedules
- Calibration procedures

End of this Section

H. Schedule of Pricing

Security Systems - 6678 HKKC2 Data Centre **Date: 29 Apr 2022** **prepared by SMW**

Item	Description	Approved Brand	Brand/Model Offered	Country of Origin	Qty	Unit	Equipment Unit Cost (HKD)	Installation Cost (HKD)	Total(HKD)
	Schedule of Pricing No. 1 - and Vendor List								
	SECURITY EQUIPMENT **Tender Adddendum 1**								
a	**Portable Metal Detector**								
	Walk Through Metal Detector, Passage height: 2050 mm , Passage width: at least 720 mm	Garrett or equivalent				NO			
	Remote control unit with (20 m) connection cable								
	5 hour external battery supply and automatic charger								
b	**Portable X-ray Detector**								
	Portable X-ray Detector, Dual energy levels Dual views Tunnel size shall be 640 mm (W) x 430 mm (H) Remote control unit with 20 m of connection cable 5 hour external battery supply and automatic charger	Rapiscan or equivalent				NO			
	The display unit , at least a 17inch LCD Monitor	Rapiscan or equivalent				SET			
	Non Powered Roller Conveyor, stainless steel 800mm width, 830mm height and 1000mm long.	Custom Made				SET			
c	**Automatic Security Entry Swing doors System**								
	Two Pairs of Double Leaf Automatic Security Entry Swing doors, with Glass in between, RC3 Security standard required software and accessories, Ground Floor	Salzer, Gilgen or equivalent				SET			
	Digital control unit with touch-screen display at Guard Room	Salzer, Gilgen or equivalent				SET			
d	**Road HVM Blocker**								
	Road HVM Blocker with accessories include but not limited to Blocking Segment, internal Hydraulic System (HPU), Blocker Control System, drainage and Cable in	Avon or equivalent				NO			
	Remote Control Panel with protection cover for On/off/stop Push button	Avon or equivalent				NO			
	Traffic Lighting Control System;	Custom Made				SET			
	Long Range Card Reader	Nedap or equivalent				NO			
	Long Range Card Reader associated Tag/Booster with sticker	Nedap or equivalent			60	NO			
	Interface marshalling box to the Access Control System provided by MEP/Security Systems Subcontractor, located in G/F Security Guard Room	Custom Made				NO			

Security Systems - 6678 HKKC2 Data Centre **Date: 29 Apr 2022** **prepared by SMW**

Item	Description	Approved Brand	Brand/Model Offered	Country of Origin	Qty	Unit	Equipment Unit Cost (HKD)	Installation Cost (HKD)	Total(HKD)
	Schedule of Pricing No. 1 - and Vendor List								
	SECURITY EQUIPMENT **Tender Adddendum 1**								
e	**Security Vehicle Shutter**								
	Security Vehicle Shutter with LPS1175 SR3 Security standard with accessories include but not limited to shutter drive unit, sensor elements leaf profiles and safety brake	Hart or equivalent				NO			
	Remote Control Panel with protection cover for On/off/stop Push button	Hart or equivalent				NO			
	Interface marshalling box to the Access Control System provided by MEP/Security Systems Subcontractor, located in G/F Security Guard Room	Custom Made				NO			
f	**Fire and Security Exit Door**								
	Fire and Security Exit Door with Forced entry resistance up to RC4, door leaf thickness 72 mm. consists of an internal extremely torsion-resistant and patented construction	Salzer, Colgen or equivalent				SET			
g	Additional Items: Please indicate additional / deduction material equipment in this item. . Modification on quantity of above items without notice is not acceptable.								